

Leo Buckel

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Education

Texas A&M University

BS in Mechanical Engineering, Mathematics Minor

GPA: 3.74

College Station, TX

May 2028

Experience

Mechanical Engineering Intern at En Solución

May – Aug 2026

- Designed mounting systems and structural supports for a mobile nanobubble ozone water-treatment system, accounting for vibration and loading experienced during transportation and operation.
- Built and assembled internal trailer components, including a climate-controlled room for electrical equipment, ventilation systems, and supports for pumps, bulkheads, and water hoses.
- Supported the scale-up of a proof-of-concept irrigation water-treatment system into a self-contained mobile unit capable of treating up to 300 gallons of water per minute.

Projects

Mechanical Linkage Football Thrower

Oct – Nov 2025

- Coordinated with a team of 3 to model a Chebyshev lambda linkage into a football player in SolidWorks, creating a motion like throwing a football.
- Fabricated a wooden prototype using a laser cutter and a 3D-printed version with print-in-place bushings to minimize friction in the joints.
- Integrated a hand-crank to actuate the linkage, iterating designs to best imitate a natural throwing motion and smooth feel.

3-Way Hi-Fi Speaker Build

Summer 2025

- Modeled 3-way speakers in Fusion 360, integrating mechanical and aesthetic design while considering material and structural constraints.
- Tested enclosures in WinISD, implementing port tuning and frequency analysis to achieve an F35 Hz.
- Manufactured speakers with resonance suppression, employing damping techniques to reduce unwanted vibrations and distortion.
- Soldered and wired crossover network, ensuring proper frequency distribution to drivers and reliable performance for 150-watt speakers.

Arduino Roulette Table

Summer 2024

- Constructed an interactive roulette table using Arduino microcontroller and breadboard circuit.
- Developed game logic in the Arduino IDE (C++) that tracked user inputs, LED ring animation, randomized outcomes, and game state transitions.
- Implemented button debouncing to ensure reliable user input without spurious open/close transitions.

Leadership & Activities

Freshmen in Engineering, Science, and Technology (FEST)

Oct 2024 – May 2026

- Mentoring first-year students, providing guidance and support to help navigate academic and campus life.
- Focused on professional growth through guest speakers and workshops covering career development, leadership, and academic success.

Skills & Interests

Technical: SolidWorks, Onshape, Python, Soldering, and Woodworking

Language: Fluent in German and English.

Interests: Band, Top Rope Rock Climbing, Volleyball, Piano, Cooking, and Soccer.